

BEMS FARMS — COMPLETE ORDER TO DELIVERY WORKFLOW

Comprehensive Technical & Operational Architecture Blueprint • From Client Checkout to Doorstep Handover, Item-by-Item Verification & Dispute Resolution

Target Audience: Engineering, Dispatch, Store Operations, Mobile Dev	System Core: Node.js / Express / PostgreSQL	Status: Production Active (v2.0)
Client Channels: Web, Mobile App, Chef Bems AI	Dispatch Mechanism: Automated GPS Proximity (Haversine)	POD Standard: Item-by-Item Photo Snapshot

1. Executive Summary & Core Topology

The Bems Farms logistics lifecycle guarantees reliable, transparent, and auditable farm-to-doorstep food delivery. The architecture synchronizes four synchronized layers: the **Client Storefront / Mobile App**, the **Chef Bems AI Culinary Assistant**, the **Admin & Warehouse Operations Hub**, and the **Driver Mobile Application**. Every order flows through distinct stages with strict geocoding verification, payment reconciliation, inventory reservation, proximity driver matching with 10-minute fallback timers, live GPS streaming, item-by-item photo capture, and instantaneous commission wallet credit.

Stage	Responsible Actor	Key System Trigger / API Endpoint	Status Mutation
1. Checkout & Payment	Customer / Chef Bems AI	POST /api/orders POST /api/orders/checkout-intent	confirmed / order_placed
2. Store Packing	Store Admin / Kitchen Staff	Packing invoice printed; warehouse stock deduction	packaging / packed_ready
3. Auto-Dispatch	Proximity Dispatch Engine	Haversine nearest driver match (10-min timeout)	assigned (driver_assigned)
4. Store Pickup	Assigned Driver	PATCH /api/driver/deliveries/:id/status (en_route)	en_route (in_transit)
5. Live Transit & GPS	Driver Device & Background Service	POST /api/driver/location (Every 10-15s)	is_on_delivery=true (Locked Unavailable)
6. Doorstep Arrival	Driver at Customer Location	PATCH /api/driver/deliveries/:id/status (arrived)	arrived (driver_arrived)
7. Item Verification & POD	Driver & Customer	Item-by-item snapshot upload: proof_photos	Physical goods verification
8. Delivery Completion	Customer & Driver Confirmation	PATCH /api/orders/:id/confirm PATCH /api/driver/deliveries/:id/status	delivered (Wallet Credited)

2. Phase 1: Customer Order Placement & Checkout

The order journey begins on the **Customer Web Storefront**, the **Customer Mobile App**, or through conversational recipe staging with the **Chef Bems AI Assistant**. All ordering paths adhere to the following mandatory validation constraints:

- Mandatory Verified Delivery Coordinates:** Customers must select a geocoded address from live OpenStreetMap search suggestions or drop an exact pin using the interactive map modal (VerifiedLocationModal). The system requires valid latitude and longitude before checkout initialization.
- Cart Staging & Inventory Reservation:** When ordering via Chef Bems AI, ingredients are staged via POST /api/ai/cart/webhook and claimed on page reload. Live stock checks prevent over-ordering.
- Server-Side Price Recomputation:** Client-supplied prices and discounts are never trusted. The server recalculates totals directly against the products table and active coupon rules in a Postgres transaction lock.

- **Payment Gateway Authorization:** For Monnify/Card/Transfer orders, a checkout_intent is locked. Monnify transactions are verified server-to-server before committing the order record.
- **Cash on Delivery (COD) Support:** For COD orders, the order is created with status pending and payment status unpaid, with the payment amount collected physically upon delivery.

3. Phase 2: Store Processing, Packing & Warehouse Checkout

Once payment is confirmed, operations transition to the store warehouse or kitchen fulfillment team:

- **1. Order Receipt & Sales Invoice:** The Store Admin receives the new order in the Admin Dashboard, inspects the line items, and prints the official packing and sales invoice.
- **2. Status Transition to Packaging:** The order status moves to packaging / packed_ready.
- **3. Warehouse Inventory Checkout:** The Delivery Manager hands the printed invoice and physical goods to the Salesperson, who verifies payment clearance and checks the items out of inventory, decrementing batch balances.
- **4. Dispatch Engine Trigger:** Once marked packaged and verified, the automated proximity dispatching subsystem is immediately invoked.

4. Phase 3: Proximity Auto-Dispatch & Driver Assignment

The automated dispatch engine (dispatchEngine.js) utilizes geospatial calculations to route deliveries to the closest available driver:

- **Haversine Distance Matching:** The engine queries all approved drivers with status = 'active', driver_availability.is_available = true, and is_on_delivery = false. It computes the geodesic distance between the store warehouse (default: lat: 5.1065, lng: 7.3667) and each driver's most recent GPS coordinates in driver_locations.
- **Assignment Creation:** The closest driver is assigned, creating a deliveries record (status: 'assigned'), updating orders.status = 'driver_assigned', and dispatching a high-priority in-app alert via driver_notifications.
- **10-Minute Response Timeout & Auto-Reassignment:** If the assigned driver does not accept or reach the store within **10 minutes**, the system automatically marks the driver assignment timed out, increments their non-response tally, and re-routes the order to the next closest available driver in the zone.
- **Manual Admin Override:** Dispatch managers can manually reassign the delivery to any specific driver if required via PATCH /api/admin/deliveries/:id/reassign.

5. Phase 4: Driver Store Pickup & Live Transit

The driver navigates to the store and commences the delivery run:

- **Store Arrival & Package Collection:** The driver presents their assignment screen, collects the packaged items and invoice, and taps 'Confirm Picked Up' (PATCH /api/driver/deliveries/:id/status with status: 'en_route').
- **In-Transit Lock:** The system sets driver_availability.is_on_delivery = true and orders.tracking_status = 'out_for_delivery'. While on active delivery, the driver is automatically excluded from receiving new order dispatches.
- **Periodic GPS Streaming:** While on transit, the driver's mobile app streams GPS location pings (latitude, longitude, heading, speed) every **10–15 seconds** via POST /api/driver/location.
- **Customer Live Tracking:** The customer can view the driver moving in real-time on the map at /track-order/:trackingCode, with updated ETAs.

6. Phase 5: Doorstep Arrival & Item-by-Item Verification

Upon arriving at the customer's delivery destination, the driver performs mandatory handover checks:

- **1. Doorstep Arrival Ping:** The driver taps '**Arrived at Location**' (status: 'arrived'). The system stamps deliveries.arrived_at and triggers an immediate push notification and SMS to the customer.
- **2. Mandatory Item-by-Item Snapshots:** To ensure complete proof of delivery and prevent missing item disputes, the driver uses the Driver App camera to capture a photo of each item/bundle delivered. Photos are saved in proof_photos and item_proofs.
- **3. Customer Physical Inspection:** The customer verifies the packaged items against their order invoice in the presence of the driver.

7. Phase 6: Delivery Outcomes & Dispute Resolution

Depending on the result of the customer inspection, the delivery concludes in one of three standardized paths:

Outcome Scenario	System & Operational Actions	Financial & Inventory Settlement
Scenario A: Successful Delivery (All Goods Correct)	1. Customer taps 'Confirm Delivery' on their app (PATCH /api/orders/:id/confirm). 2. Driver clicks 'Complete Delivery' (status: 'delivered'). 3. System sets orders.status = 'delivered' and deliveries.delivered_at = NOW().	• Driver commission (e.g. ₦500 – ₦1,200) is instantly credited to drivers.total_earnings and recorded in driver_commissions. • Driver availability resets to is_on_delivery = false, ready for next dispatch.
Scenario B: Customer Unavailable (No Answer at Door/Phone)	1. Driver taps 'Customer Unavailable' (status: 'delivery_attempted'). 2. System sends urgent SMS/Push alerts & starts a 15-Minute Countdown. 3. If customer reaches out within 15 mins \$rightarrow\$ Delivery is completed. 4. If timer expires \$rightarrow\$ Attempt counter increments (max 2 attempts).	• Dispatch Manager receives alert in Admin Portal. • Admin either schedules a re-delivery window or directs the driver to return items to warehouse. • Returned items are checked back into inventory.
Scenario C: Issue / Dispute Reported (Damage, Missing Item, Defect)	1. Customer taps 'Report Issue' on their app (POST /api/orders/:id/report) and uploads photos of discrepancy. 2. Delivery Manager receives high-priority dispute ticket in Admin Portal. 3. Admin reviews customer photos against the driver's POD item snapshots.	• Full/Partial Refund: Credited to customer wallet or original bank account. • Replacement Dispatch: Store dispatches replacement immediately via another courier. • Returned damaged goods are logged under lost/damaged inventory ledger.

8. Driver Wallet & Automated Commission Payouts

Upon completing deliveries, driver earnings are managed through an automated ledger and bank payout system:

- **Real-Time Commission Accrual:** Every successfully delivered order immediately credits the driver's wallet with their per-drop commission (configured per zone and vehicle type).
- **Dedicated Inflow Virtual Account:** Each driver receives a dedicated Monnify Virtual Account (e.g. 855XXXXXXX - Monnify / Wema Bank) for direct credits and earnings tracking.
- **Self-Service Bank Withdrawals:** Drivers can request instant withdrawals from their mobile wallet (POST /api/driver/withdraw) to their personal Nigerian bank account. Withdrawals are processed and disbursed via the Admin Payouts Manager.

9. Operational Verification & Compliance Checklist

Checklist Item	Standard Requirement	Enforcement Layer
Address Geocoding	Mandatory valid GPS latitude & longitude coordinates	Frontend validateForm & Backend Zod Order Schema
Driver Availability Lock	Unverified/Suspended drivers cannot toggle online; in-transit drivers locked	PATCH /api/driver/availability & dispatchEngine
Dispatch Timeout	10-Minute non-response threshold triggers auto-reassignment to next driver	Backend Cron / Dispatch Engine Timer
Proof of Delivery (POD)	Item-by-item photo capture uploaded to proof_photos	Driver Mobile App Camera & Multipart KYC/POD Endpoint
Double Confirmation	Both customer and driver confirmation required for clean status closing	/orders/:id/confirm & /driver/deliveries/:id/status